7 **(a)** Find $\int x^3\left(15x+\frac{11}{\sqrt[3]{x}}\right)\mathrm{d}x$. **[5]**

**(b)** Show that $\int_0^8 x^3\left(15x+\frac{11}{\sqrt[3]{x}}\right)\mathrm{d}x = a\times 2^{11}$, where $a$ is a positive integer to be determined. **[3]**

**7(a)**

**7(b)**